What explains temperature differences in Heilbronn?

Oct 28, 2026 · @Ryuta

Key findings

Three factors explain most of the temperature differences in our measurements: industrial land north of the centre, open land and river corridors on the edges, and tree canopy. All differences are small in absolute terms (tenths of a kelvin), because they are measured within one trip, but they are statistically clear and match the city's own 2017 climate analysis.

1. **Industrial north is warmest, open land in the south-west is coolest.** Bike cells in the industrial corridor between the centre and the sewage works averaged +0.18 K; cells south-west of the centre (Neckar meadows, open land) averaged −0.13 K. In the evening the gap widens to 0.52 K (+0.33 K vs −0.19 K).
2. **The edge of the city cools first in the evening.** Controlling for tree cover, every km from the centre lowers the anomaly by 0.10 K in the evening (95% CI 0.07–0.13), but only by 0.02 K at midday. Cells 5–8 km out were 0.29 K cooler in the evening.
3. **More tree canopy means cooler, more humid air.** Each extra 10% of crown cover within 100 m lowers the anomaly by 0.07 K (95% CI 0.04–0.10) and raises relative humidity by 0.4 %-points. The effect holds at morning, midday and evening, and inside the urban core alone.

One methodological finding matters for anyone using the walking-sensor data: sensors walking side by side read 1–2 K apart, so raw comparisons between devices are not valid (see Data and method).

Data and method

The analysis compares each measurement with the rest of its own trip, so weather and time of day cancel out and only place-to-place differences remain.

Dataset	What it contains	Used here
Bike sensor (OSEM_Bike_Data_cleaned.csv)	54,979 readings, 5 Aug–14 Sep 2026, temperature + humidity every ~5 s	49,686 readings in 174 trips inside Heilbronn

Dataset	What it contains	Used here
Walking sensors (alles_kid_temp_clean.csv)	61,410 readings from up to ~8 devices, 1–3 Sep 2026	36,368 readings from the 2 Sep city walk
Baumkataster Heilbronn 2026	43,031 city-registered trees with crown diameter (dl-de/by-2.0)	Tree-crown cover around each cell
Gesamtstädtische Klimaanalyse 2017	Official climate stations (2015/16), climatope map	Centre point and station comparison

Removing weather and time of day. Each bike trip was split at gaps over 5 minutes, and a straight-line time trend was removed from each trip. This is the same correction the city's climate analysis applied to its e-bike measurement runs, to remove the general cooling during the night (Stadt Heilbronn 2017). The result is an anomaly in kelvin: how much warmer or cooler a place was than the trip's trend. Anomalies were averaged in 100 m grid cells; only cells visited on at least 2 trips were kept (1,045 cells).

The walking-sensor offset problem. The walking data has no device ID. Rebuilding each device's track from positions showed that sensors walking side by side differed by 1–2 K, larger than the real spatial signal. A first, naive comparison against the all-device mean produced a misleading result (apparently warmer away from the centre) that came from which device walked where. The fix: each device track was cut into 10-minute pieces and detrended separately, so a constant device offset cancels out. Only local contrasts within a ~10-minute walk remain.

Tree canopy. For each cell, the crown areas of all registered trees within 100 m (50 m for walking data) were summed and divided by the circle area, giving percent crown cover. The cadastre's own readme notes that some trees not maintained by the city carry a default 2 m crown, so cover is underestimated in places.

Statistics. Weighted least-squares regression of cell anomaly on crown cover and distance from the centre (Klimaanalyse station St.1, Innenstadt), with robust (HC3) standard errors. Both variables are needed together, because street trees are concentrated in the centre (Spearman $\rho = -0.67$ between canopy and distance).

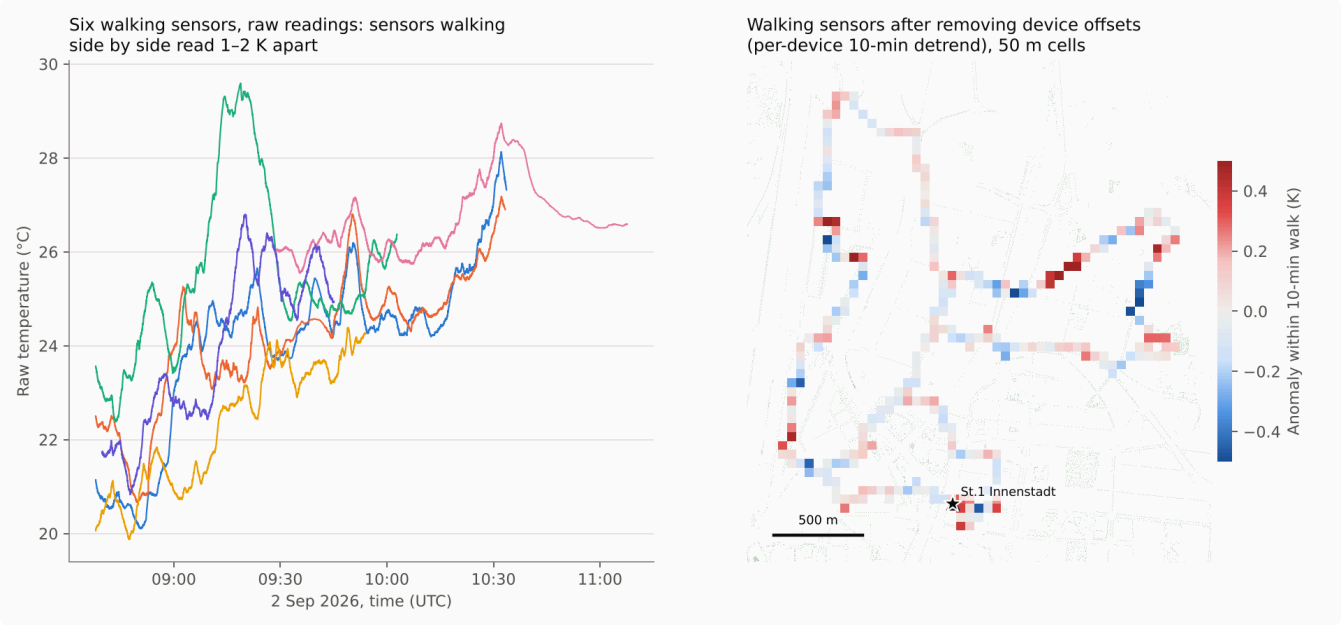


Figure 1. Left: raw readings of six walking sensors on 2 Sep 2026. Right: local anomalies after per-device 10-minute detrending, 50 m cells, on registered trees (green).

Where it is warm and where it is cool

The industrial corridor north of the centre is the warmest part of the measured network; the Neckar meadows and open land to the south-west are the coolest.

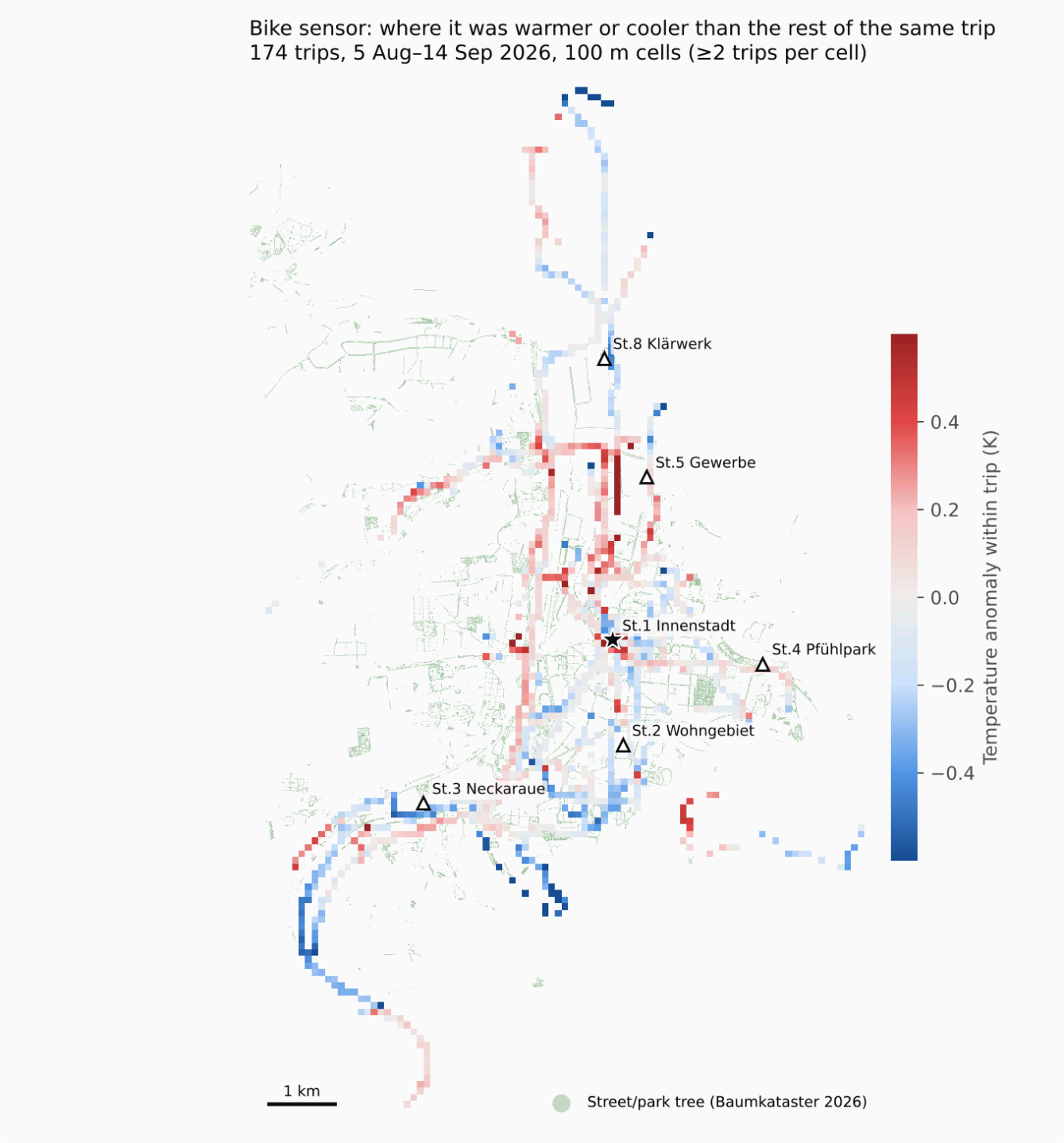


Figure 2. Bike temperature anomaly per 100 m cell (red = warmer than the rest of the trip, blue = cooler), all 174 trips. Triangles: official Klimaanalyse stations; star: city centre (St.1).

Zone (bike cells)	All trips	Evening (16–21 UTC)	Mean crown cover
Industrial corridor north of centre (111 cells)	+0.18 K	+0.33 K	4%
Inner city, within 700 m of St.1 (94 cells)	0.00 K	+0.03 K	10%

Zone (bike cells)	All trips	Evening (16–21 UTC)	Mean crown cover
South-west: Neckar meadows and open land (234 cells)	−0.13 K	−0.19 K	2%

The north–south-west difference is 0.31 K over all trips and 0.52 K in the evening (both $p < 0.001$, Welch t-test). The inner city is not the hottest zone, even though the city's climate analysis expects it to be; it also has the most street trees (10% crown cover), which is consistent with the canopy effect in the next section.

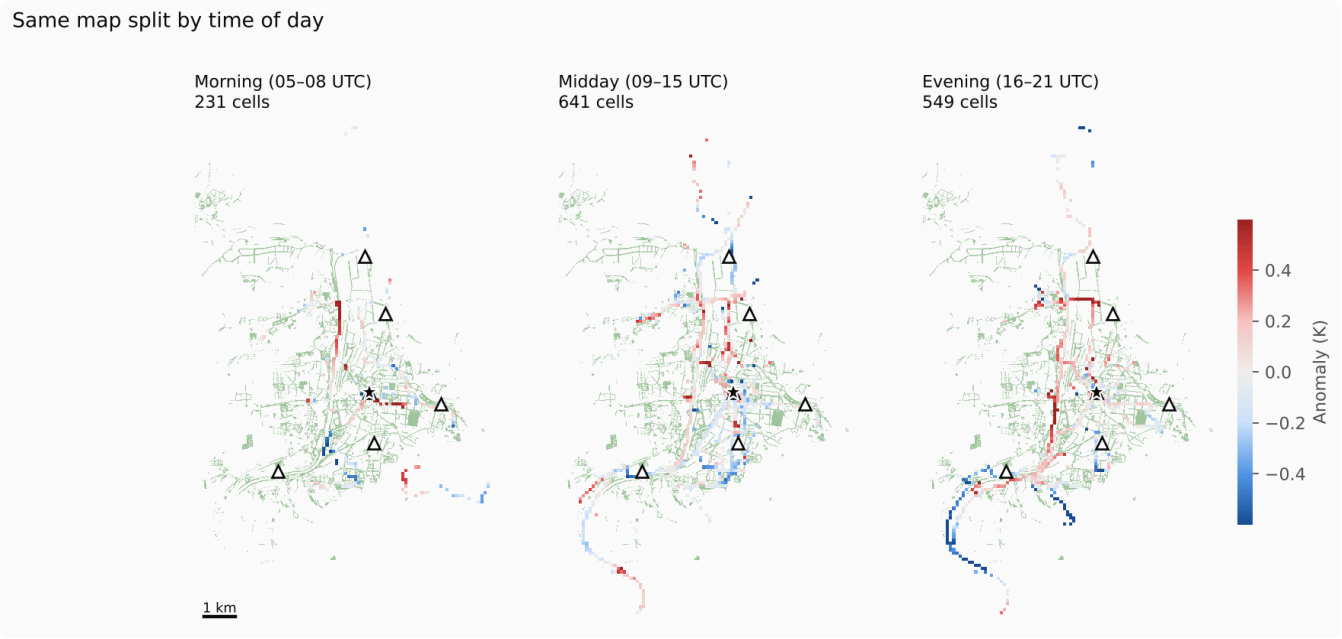


Figure 3. The same anomalies split by time of day. Times are assumed to be UTC (openSenseMap's storage convention); add 2 hours for local summer time.

The contrast between city and edge depends strongly on time of day.

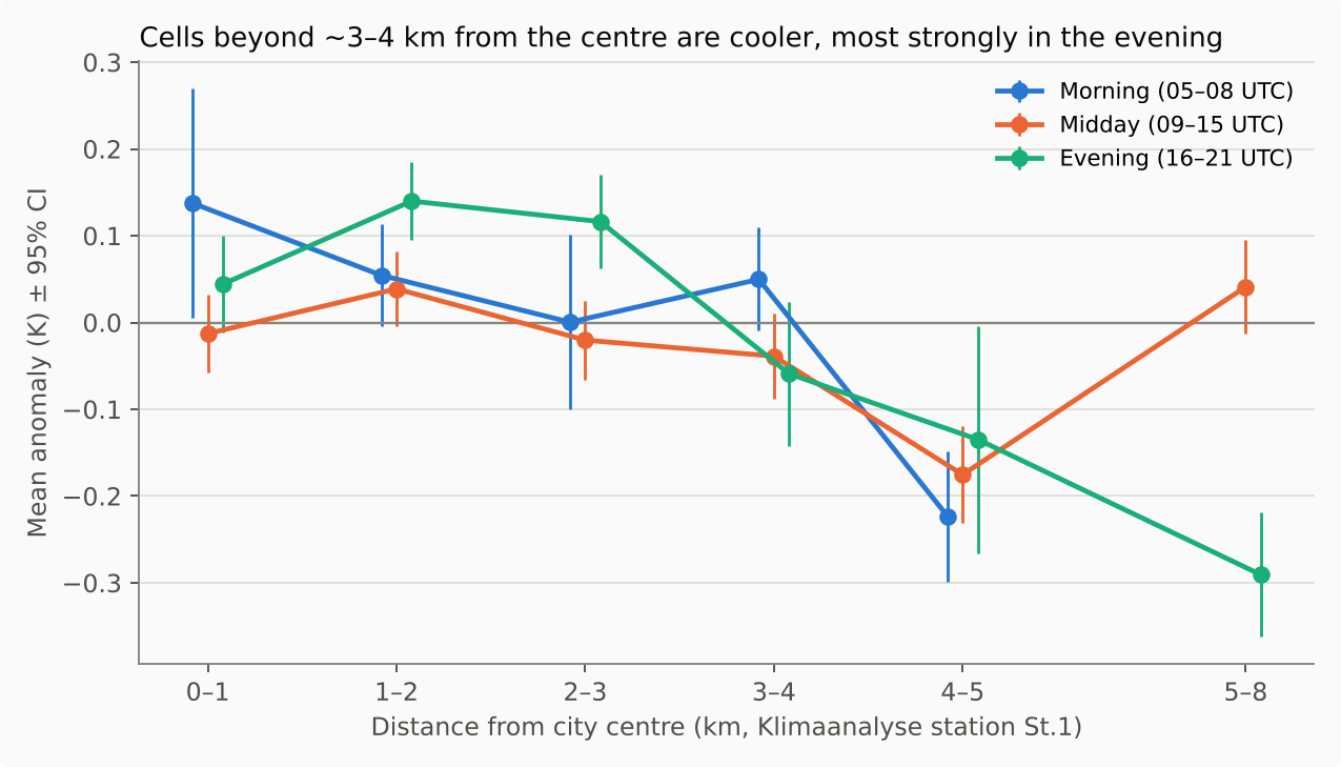


Figure 4. Mean anomaly by distance band from the centre, ± 95% confidence interval.

Time of day	Change per km from centre (95% CI)	Cells
Morning (05–08 UTC)	−0.06 K (−0.10 to −0.03)	231
Midday (09–15 UTC)	−0.02 K (−0.03 to −0.00)	641
Evening (16–21 UTC)	−0.10 K (−0.13 to −0.07)	549

In the evening, cells 1–3 km from the centre were +0.12 to +0.14 K and cells 5–8 km out were −0.29 K. At midday the profile is almost flat. Within 3 km of the centre the distance effect largely disappears (−0.03 K/km, p = 0.12, all trips); the gradient comes from the transition to the open edge of the city.

Tree canopy

Once distance from the centre is held constant, places with more tree crown cover are measurably cooler and more humid, at every time of day.

The raw correlation points the wrong way (Spearman ρ = +0.11): cells with almost no registered trees are mostly at the city edge, which is cooler for other reasons. The cadastre covers city street and park trees, not farmland or private gardens, so "few trees" there means "outside the street network", not "bare". Controlling for distance removes this confounding.

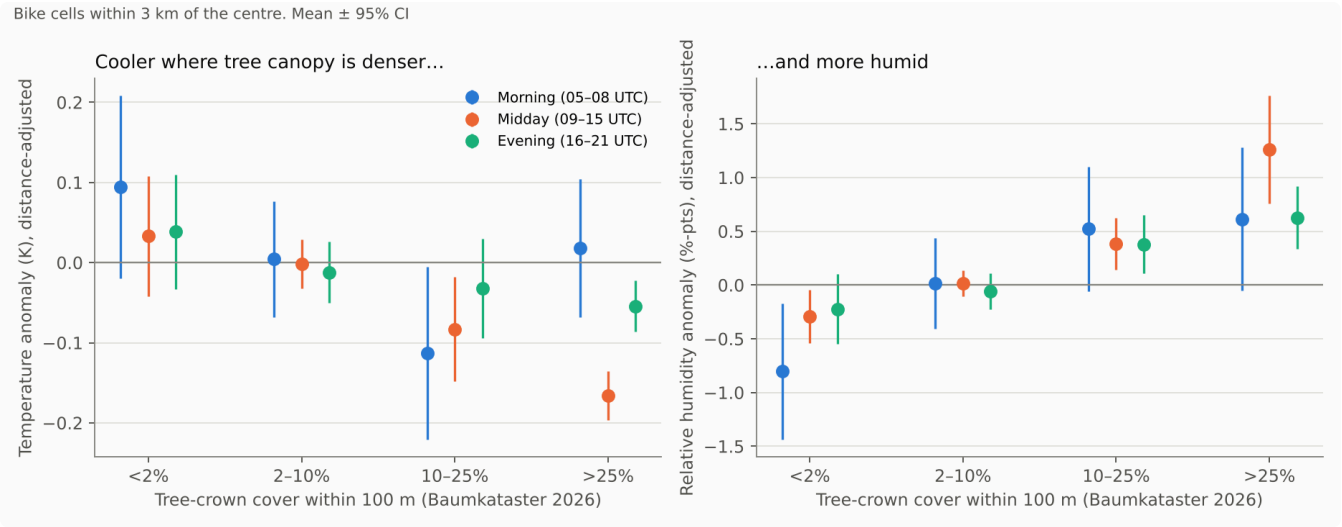


Figure 5. Distance-adjusted anomalies by crown-cover class, bike cells within 3 km of the centre, ± 95% CI. Classes with fewer than 4 cells are hidden.

Time of day	Temperature per +10% crown cover (95% CI)	Humidity per +10% crown cover
All trips	−0.07 K (−0.10 to −0.04)	+0.4 %-pts
Morning	−0.10 K (−0.16 to −0.04)	+0.7 %-pts
Midday	−0.06 K (−0.10 to −0.02)	+0.4 %-pts
Evening	−0.08 K (−0.12 to −0.03)	+0.4 %-pts

All temperature and humidity effects are significant at $p < 0.01$. Restricting to cells within 3 km of the centre gives a similar effect (−0.09 K per +10%, $p < 0.001$), so the result is not driven by the city edge. In the inner core, the 5 cells with more than 25% crown cover were about 0.17 K cooler at midday.

The walking sensors did not show a canopy effect (−0.02 K per +10%, $p = 0.20$). That test is weak: one 2-hour walk around midday, only local contrasts within a 10-minute walk, and few cells with dense canopy (10 cells above 25%).

Comparison with the official climate analysis

Our measurements broadly agree with the city's 2017 climate analysis: places it classes as open land or river corridor were cooler, places it classes as inner city or industrial were warmer.

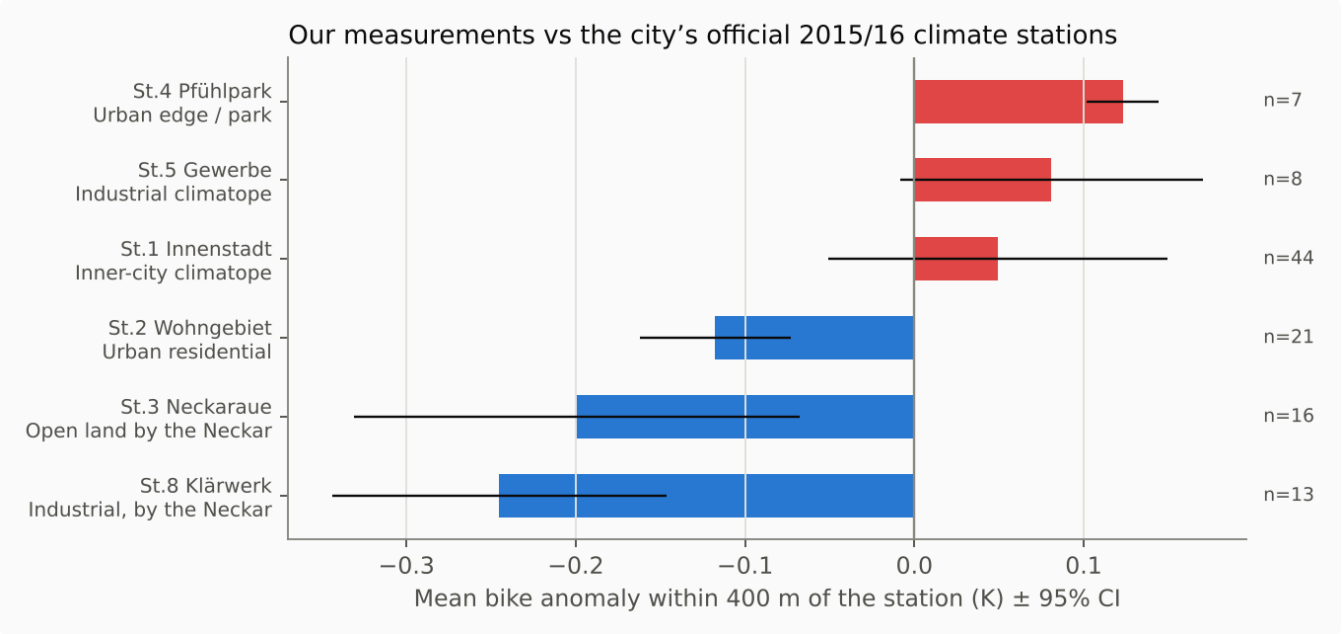


Figure 6. Mean bike anomaly in cells within 400 m of each Klimaanalyse station, ± 95% CI; n = number of cells. Station positions converted from the report's Gauss-Krüger coordinates.

- **Cool, clearly below zero:** Klärwerk (−0.25 K) and Neckaraue (−0.20 K), both beside the Neckar, and the residential station St.2 (−0.12 K). The analysis describes the Neckar and its harbours as water climatopes that act as ventilation corridors, and the open land as a source of cold air (Stadt Heilbronn 2017).
- **Warm, but not significant:** Innenstadt (+0.05 K) and Gewerbe (+0.08 K); their confidence intervals include zero. The analysis classes both inner city and industrial areas as heat-burden zones.
- **Unexpected:** the area around the Pfühlpark station was warm (+0.12 K), but only 7 cells were sampled and the 400 m circle includes streets outside the park. Treat it as inconclusive.

The analysis also found that cold air from the surrounding land does not reach deep into the built-up area, because building edges block it and there are no ventilation corridors into the city. That fits Figure 4: the cooling appears only beyond about 3 km from the centre.

Causes

Each pattern has a physical mechanism documented in the literature and supported by our data; the strength of the evidence differs.

Pattern in our data	Likely cause	Our evidence	Supporting source	Strength
Industrial north warmest (+0.18 K; +0.33 K evening)	Large sealed and built surfaces store heat by day and release it in the evening; little vegetation	Warmest zone has only 4% crown cover; gap grows in the evening	Oke 1982; Klimaanalyse classes the Kanalhafen, Neckar and Osthafen industrial areas as heat-burden zones (Stadt Heilbronn 2017)	Strong (map pattern, 111 cells, $p < 0.001$)
City edge cooler, mostly in the evening (−0.10 K/km)	Open land cools quickly after sunset; built-up areas release stored heat slowly, so the heat-island contrast builds in the evening	Distance effect 5× larger in the evening than at midday	Oke 1982; the Klimaanalyse found the inner city cools least at night	Strong ($p < 0.001$), but only for cells beyond ~3 km
More tree crown cover → cooler (−0.07 K per +10%) and more humid (+0.4 %-pts)	Shading of the ground and evapotranspiration from leaves	Effect at all times of day; humidity rises where temperature falls, the signature of evaporation	Bowler et al. 2010; Ziter et al. 2019	Moderate: robust statistically, small in size
Neckar corridor and meadows cooler (−0.20 to −0.25 K near stations)	Water and open meadows act as ventilation and cooling corridors	Two river-side stations clearly below zero	Klimaanalyse: water climatopes act as ventilation corridors (Stadt Heilbronn 2017)	Moderate: water and open land cannot be separated without land-use data

Why the tree effect is small. Ziter et al. found that cooling becomes large only when canopy cover exceeds about 40% at block scale (Ziter et al. 2019). Only 10 of our 1,045 cells had more than 25% registered crown cover, so Heilbronn's streets mostly sit below the range where trees cool strongly. This is consistent with a small but real effect.

Why the inner city is not the hottest zone. The inner city has the most street trees in our sample (10% crown cover vs 2–4% elsewhere), and the canopy effect predicts roughly 0.04–0.06 K of cooling from that difference. That partly offsets its dense building. The evening data also stop at 21 UTC (23:00 local), so the late-night hours, when the heat island is usually strongest, are not covered (Oke 1982).

What these data cannot show. Sealed surface, building height and distance to water were not measured here, so their share is inferred from location and from the Klimaanalyse, not tested directly. The next step would be adding imperviousness or building data (for example Copernicus Imperviousness Density or LGL LoD2 buildings) as further regression variables.

Limitations

- **Relative, not absolute.** Anomalies compare places within one trip. A trip that stays in one district shows no difference, so large-scale contrasts are underestimated.
- **Moving sensors lag.** At ~5 m/s the sensor needs time to adjust, which smears the signal over roughly 100 m; this is why 100 m cells were used.
- **Sun on the sensor.** Unshielded low-cost sensors can read too warm in direct sun, which may exaggerate daytime differences between sunny and shaded streets. Fenner et al. (2017) list non-standard measurement locations among the open quality issues of crowdsourced temperature data.
- **Timezone assumed.** Bike timestamps carry no zone; UTC was assumed. The hourly mean temperature peaks at 12–16 UTC (14–18 local), which supports that.
- **Tree data incomplete.** The cadastre holds city-registered trees only. Private gardens and forests are missing, and some trees carry a placeholder 2 m crown.
- **Correlation, not experiment.** Canopy, sealing and building density vary together. The regression separates canopy from distance, but not from variables that were not measured.

References

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5. Stadt Heilbronn (2026). *Baumkataster des Stadtkreises Heilbronn*, Stand 01.01.2026, Datenlizenz Deutschland – Namensnennung 2.0. Offene Daten Heilbronn
6. Ziter, C. D., Pedersen, E. J., Kucharik, C. J., & Turner, M. G. (2019). Scale-dependent interactions between tree canopy cover and impervious surfaces reduce daytime urban heat during summer. *Proceedings of the National Academy of Sciences*. https://doi.org/10.1073/pnas.1817561116

Measurement data: OSEM_Bike_Data_cleaned.csv (openSenseMap bike sensor) and alles_kid_temp_clean.csv (walking sensors), provided by the author. Analysis in Python (pandas, geopandas, statsmodels).